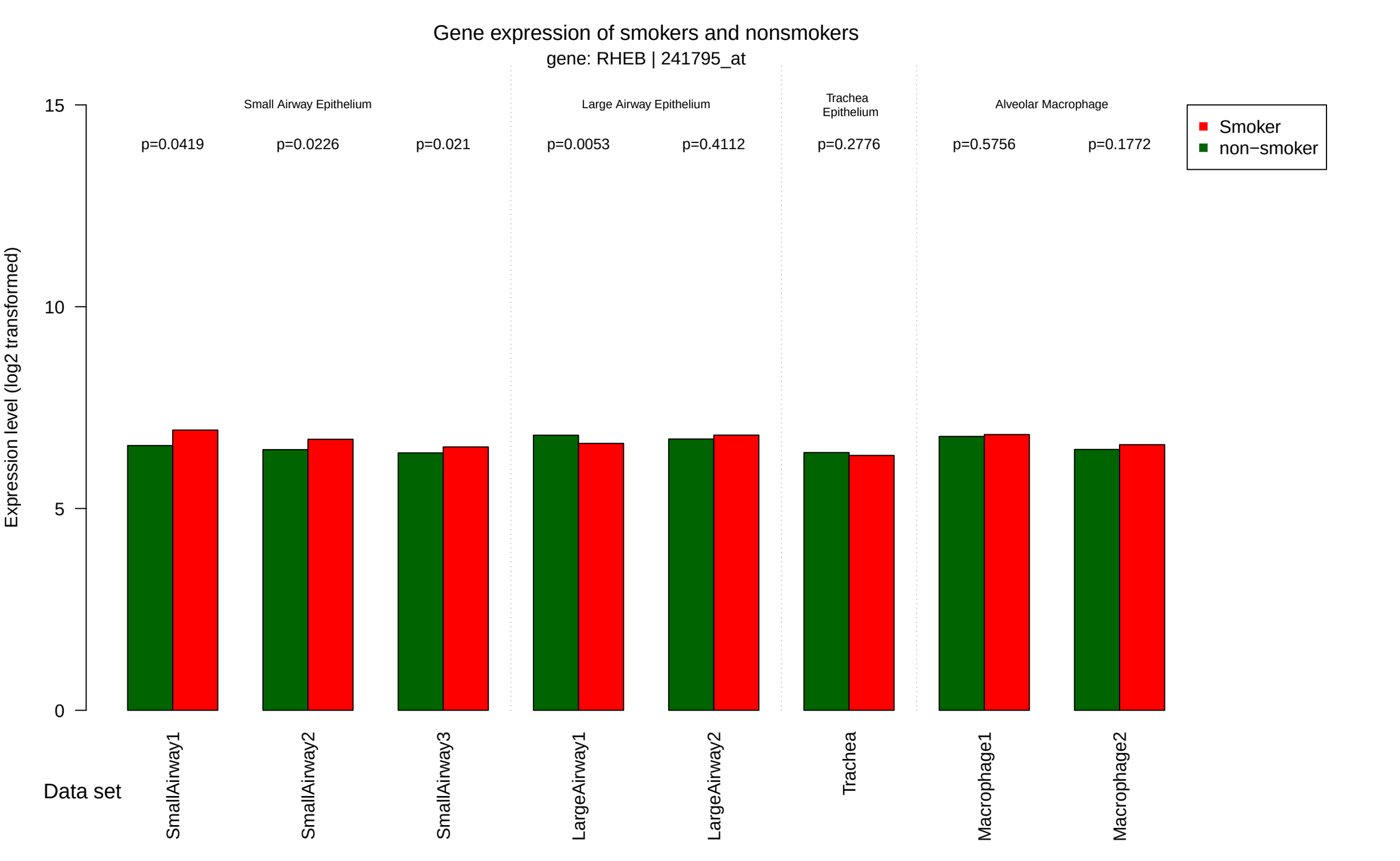
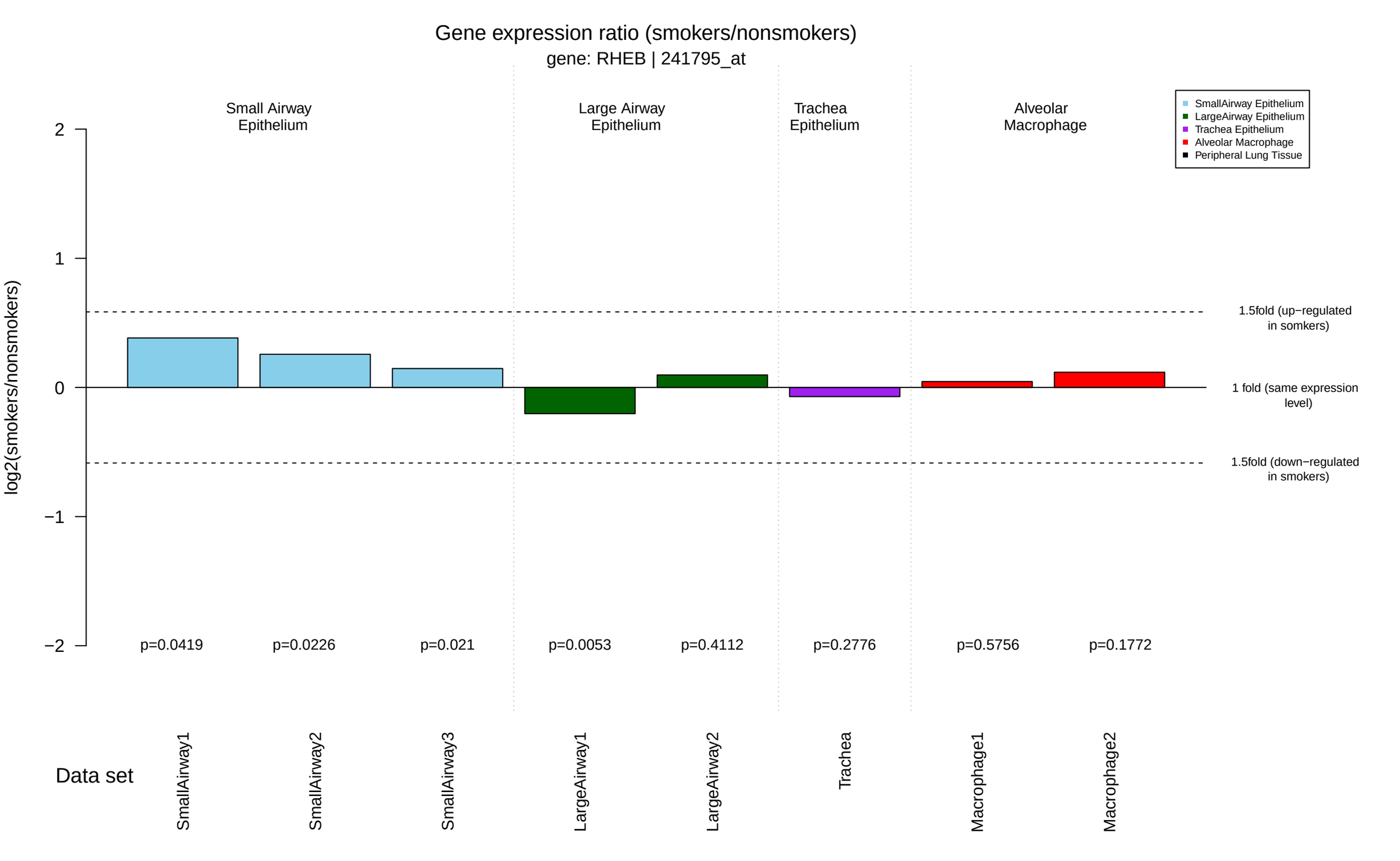
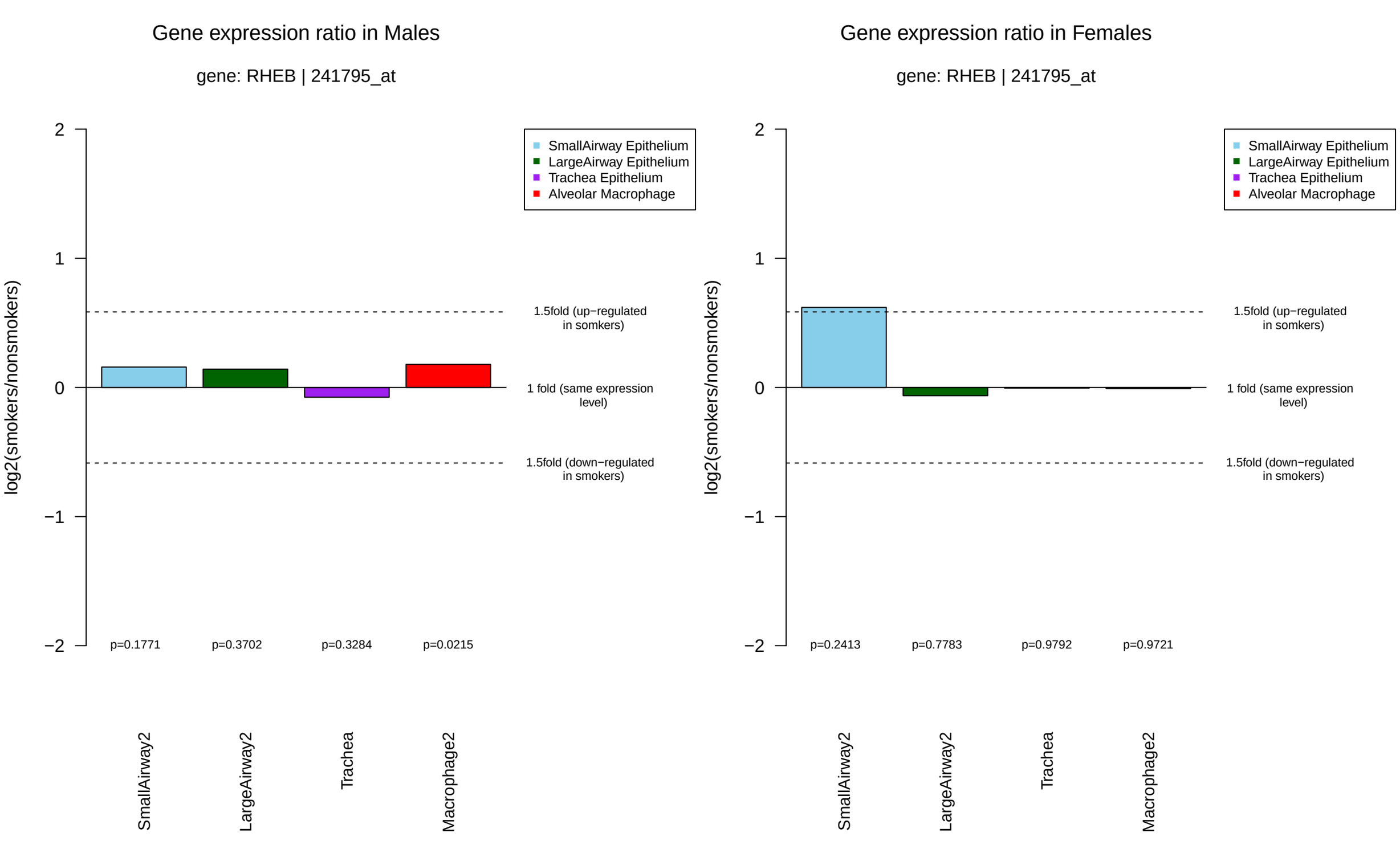
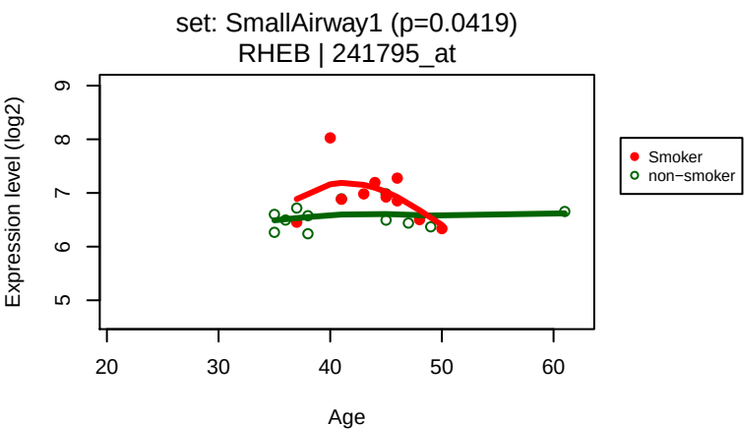
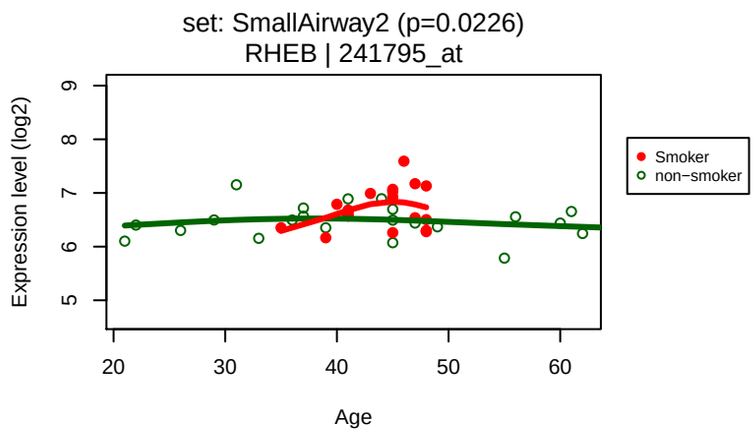
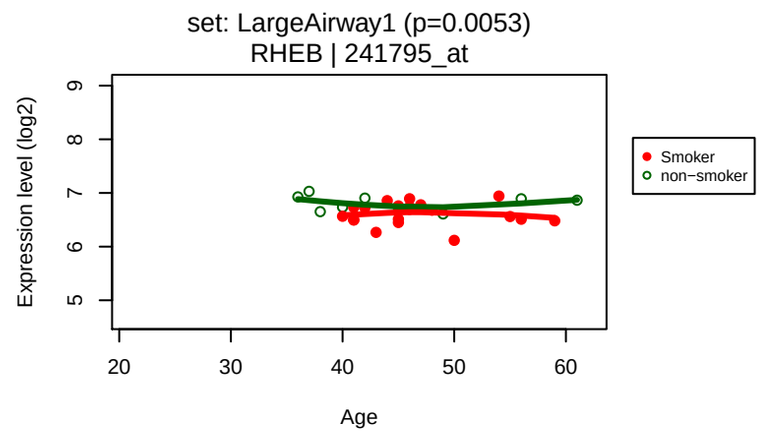

Gene expression of smokers and nonsmokers
gene: RHEB | 241795_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0419
p=0.0226
p=0.021
p=0.0053
p=0.4112
p=0.2776
p=0.5756
p=0.1772
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

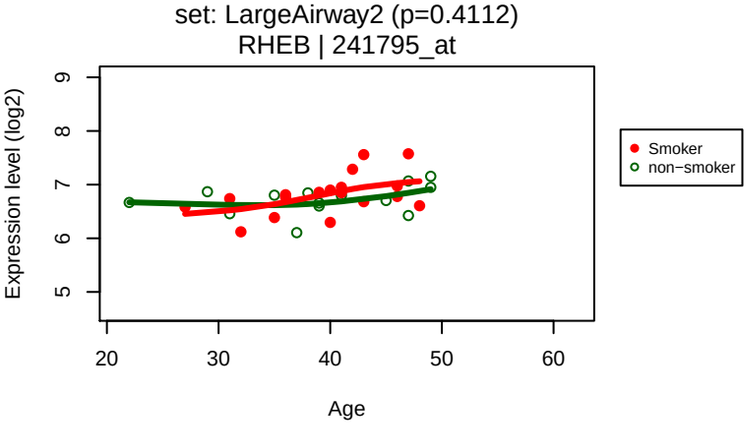

Gene expression ratio (smokers/nonsmokers)
gene: RHEB | 241795_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0419
p=0.0226
p=0.021
p=0.0053
p=0.4112
p=0.2776
p=0.5756
p=0.1772
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

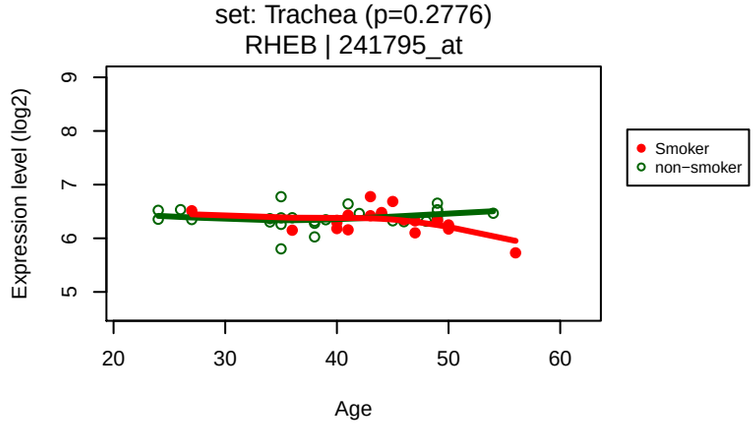

Gene expression ratio in Males
gene: RHEB | 241795_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1771
p=0.3702
p=0.3284
p=0.0215
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RHEB | 241795_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2413
p=0.7783
p=0.9792
p=0.9721
SmallAirway2
LargeAirway2
Trachea
Macrophage2

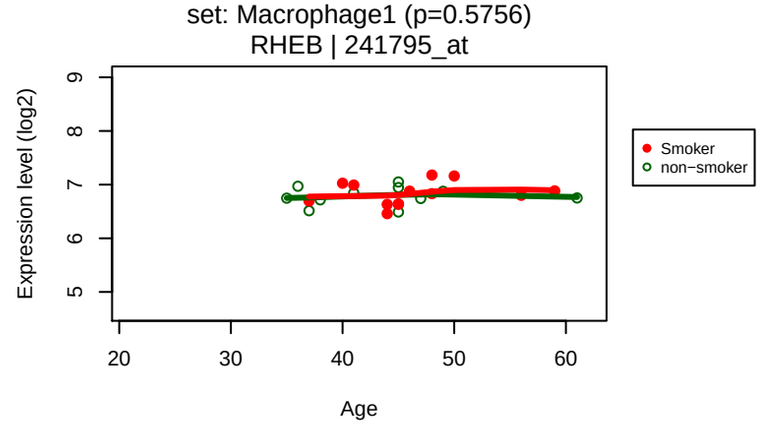

set: SmallAirway1 (p=0.0419)
RHEB | 241795_at
Expression level (log2)
Age
Smoker
non−smoker

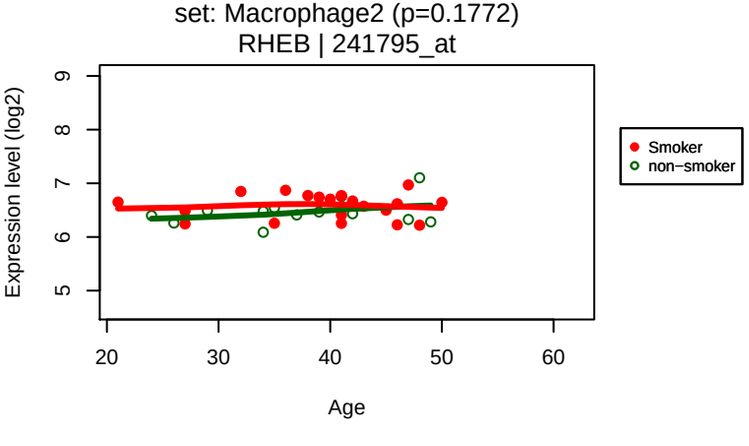

set: SmallAirway2 (p=0.0226)
RHEB | 241795_at
Expression level (log2)
Age
Smoker
non−smoker

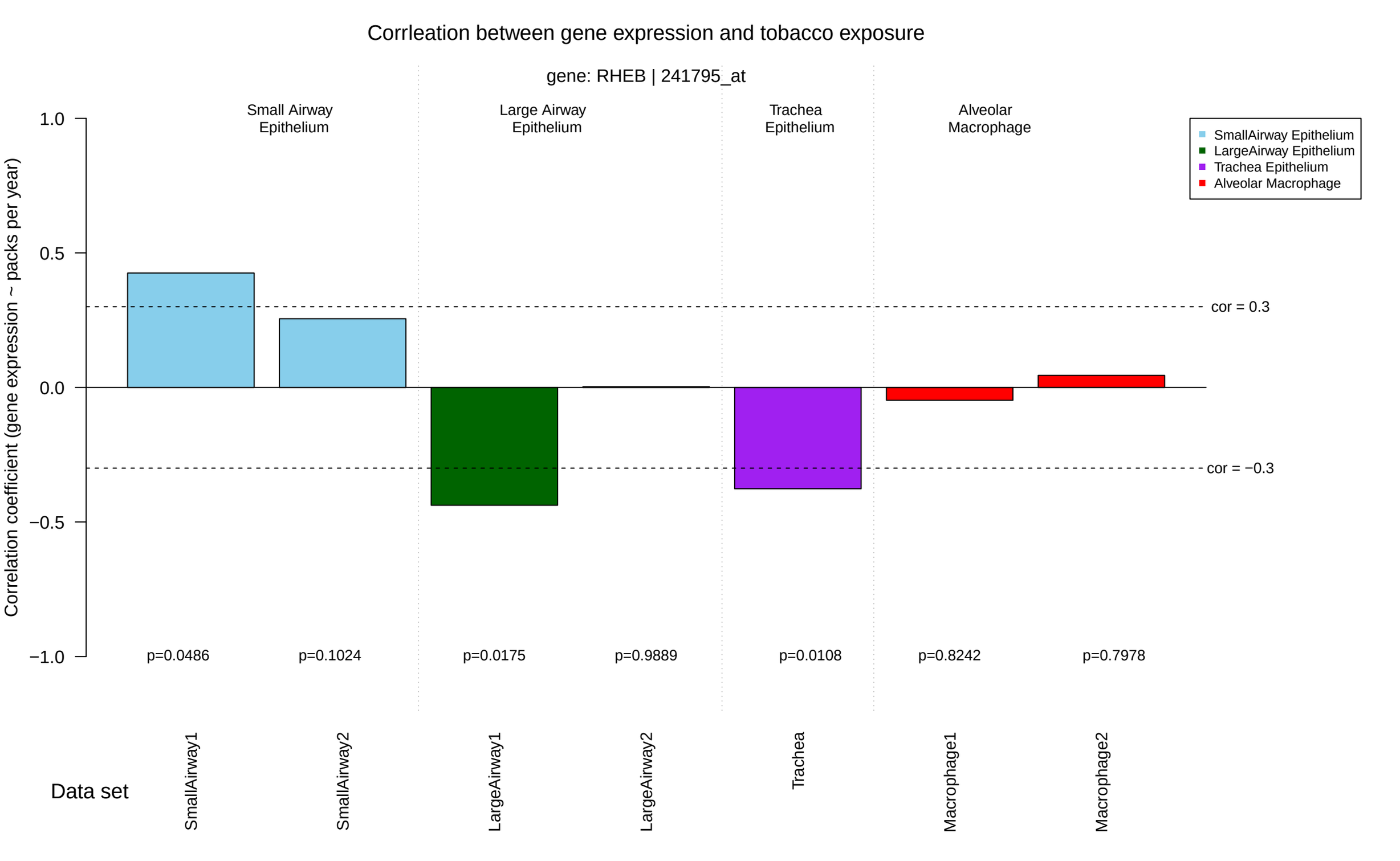

set: LargeAirway1 (p=0.0053)
RHEB | 241795_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4112)
RHEB | 241795_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2776)
RHEB | 241795_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5756)
RHEB | 241795_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1772)
RHEB | 241795_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RHEB | 241795_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0486
p=0.1024
p=0.0175
p=0.9889
p=0.0108
p=0.8242
p=0.7978
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2